Offline functionality test report

Adventure Timeline, GitHub issue #12 · 10 November 2026

Summary

The app was tested on a real phone with Wi-Fi switched off, covering everything that could be exercised safely: browsing, creating and editing a trip, capturing photo, video and audio, incidents, sharing and export, settings, a full 3.7 GB backup and a restore. Every feature worked offline. Features that depend on the internet (update check, place search, place names) fail with clear messages or fall back to coordinates.

Three bugs were found that are not offline-specific:

1. A photo or video taken with the in-app camera is silently lost if Android reclaims the app while the camera is open. **Fixed** in `MainActivity.kt` and verified (uncommitted).
2. Deleting a trip leaves its photos, videos and voice notes on disk, so storage cannot be reclaimed. **Not fixed**; reported below.
3. Rotating the phone wiped text typed into forms and the Share message, closed dialogs, and orphaned a voice note that was being recorded. **Fixed** in `AndroidManifest.xml` and verified (uncommitted).

Test setup

- Build: debug build of the working tree on commit e463f7b, plus the Finding 1 and Finding 3 fixes (installed with adb). Version shown in app: 0.0.8-5-gda46057.
- Device: Samsung SM-N986B, Android 13, Wi-Fi only. Wi-Fi was disabled with `svc wifi disable`; the phone reported “Network is unreachable”. Wi-Fi was turned back on at the end.
- Method: app driven over adb (taps located by accessibility text, screenshots captured at each step). The phone holds the owner's real diary, so data-entry tests used one throwaway trip (“ZZ Offline Test Trip”).
- Safety: the database was copied before testing and compared at the end. Trips (21), legs (395) and route points (6,867) are identical to the start.
- Network use found by code audit: map tiles (OpenStreetMap), place names (Android Geocoder, every call wrapped so failure returns nothing), and the update check (only when tapped).

Results

Feature	Result	Observed offline
Cold start, all-trips map	**Pass**	Opens normally. Routes, waypoints and incident markers draw over cached tiles. Waypoints/Photos/Incidents toggles work.
Centre on my location	**Pass**	GPS fix works with no network; blue dot shown.
Trip stats	**Pass**	Totals, per-transport bars and per-year bars load from local data.
Have I been here before?	**Pass**	Works. It looks for logged photos, videos and incidents near the current location (not trips), so “none near home” is correct.
Trip list, sort by rating, title search, Near me	**Pass**	All work. Near me used GPS and found 13 trips within 5 km.
Trip detail: Map, Legs, Media, Notes, Incidents, Share tabs	**Pass**	All load; thumbnails render.
Create trip, edit notes, date	**Pass**	Trip saved and edited offline.

picker, rating		
Add leg, edit leg (transport, difficulty, enjoyment)	**Pass**	Saved; shown as “Difficulty 4/5 · Enjoyment 5 stars”.
Capture photo (camera)	**Bug found**	Works normally, but a photo was silently lost once when the app was reclaimed while the camera was open. Reproduced and fixed. See Finding 1.
Record video; record voice note	**Pass**	Both saved and shown in Media (1 photo, 1 video, 1 audio).
Media viewer: photo, video, audio playback	**Pass**	Works. The photo shows raw coordinates instead of a place name (place lookup unavailable offline), which is the intended fallback. Audio played 0:04 / 0:14.
Log incident (type, severity, people, details); delete incident	**Pass**	Saved offline, listed on the trip and the Incidents tab; deleted.
Screen rotation: New trip, Add leg, incident form, Share message, Settings dialog	**Bug found**	Typed text and the open dialog were lost on rotation. Fixed and re-tested: all keep their state. See Finding 3.
Screen rotation during a voice recording	**Bug found**	Before the fix the screen reset to “Start recording” and the recording was orphaned. After the fix it keeps recording and saves normally (“1 audio”).
Camera photo with the phone rotated by hand	**Pass**	The owner turned the phone sideways with the camera open and took a photo (with the fixes installed). The photo attached to the trip correctly, with its location, and the app stayed on the Media tab.
Screen rotation in the photo viewer	**Pass**	Stays on the same item (3 / 7) in landscape and back.
Delete trip	**Bug found**	Trip rows deleted, but its media files stay on disk. See Finding 2.
Share as Post, Picture, PDF, GPX, KML	**Pass**	Every format builds and opens the system share sheet. Picture and PDF render correctly. GPX/KML are valid but empty for a trip with no route.
Export trip data (zip)	**Pass**	Built in under 2 seconds; valid trip.json plus the photo.
Share app; Send feedback	**Pass**	Share app opens the chooser. Send feedback opens an email to adventuretimeline@jonesie.net.nz with subject and body (version, Android) filled in. Draft discarded, nothing sent.
Settings: Theme, Home area, Backup reminder, Incident reasons, About	**Pass**	All dialogs and screens work. Theme changed to Light and restored to Dark.
Back up diary (3.7 GB)	**Pass**	Progress dialog, then “Backed up 22 trips and 10 photo/video/audio files (3.7 GB)”.
Restore from backup	**Pass**	“Restored 22 trips and 10 media files” in about 95 s. Database compared

		afterwards: trips, legs and all 6,867 route points identical.
Link companions to contacts	**Pass**	Permission denied path shows “Contacts permission is needed to match companions.” Granted path not tested.
Import from Google Timeline	**Pass**	Screen opens. An actual import was not run (no Timeline.json used).
Check for updates	**Pass**	Fails cleanly: “Couldn't check just now. You may be offline…”
Place search (Trips → Place)	**Pass**	Clear message: “Place lookup needs a network connection.”

Findings

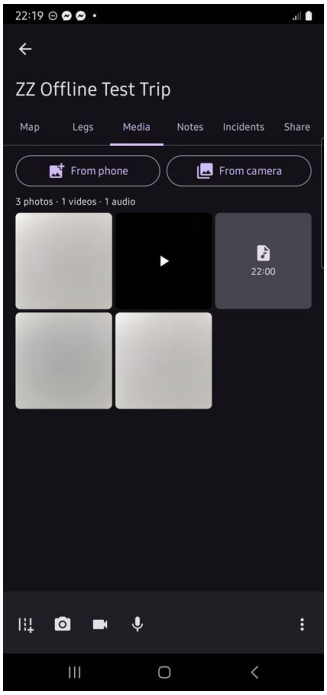
Finding 1: photo or video lost when the app is reclaimed during capture (fixed)

What happened. On the first camera use the photo never appeared in the trip (media count unchanged), after a long pause in the camera app. A second, quicker attempt worked.

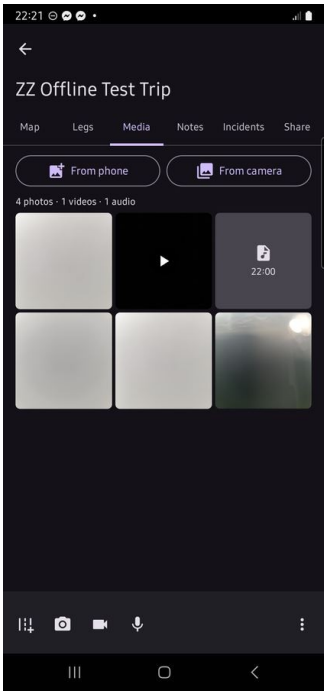
Cause. The pending photo and video URIs were held in `remember`, which is lost if Android recreates the app while the camera is in front. The camera result then arrives with no URI to attach and is silently dropped. The phone had granted camera access as “Only this time”, which Android can revoke (restarting the app) after the app has been in the background for a while; that is the likely trigger but was not proven.

Reproduction. Open the camera from a trip, take a photo, run `am kill` on the app while the camera review is showing, then accept the photo. Before the fix the app restarted on the Media tab and still showed 3 photos (photo dropped). After the fix it showed 4 photos.

Fix. `pendingPhotoUri` and `pendingVideoUri` in `MainActivity.kt` now use `rememberSaveable`. Not covered: incident-level capture in `IncidentList.kt` has the same pattern but holds a database entity that needs a small saveable wrapper.



Before fix: app killed, photo accepted. Still 3 photos.

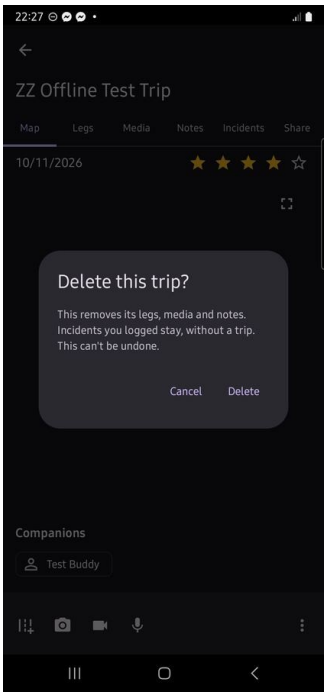


After fix: same steps. 4 photos.

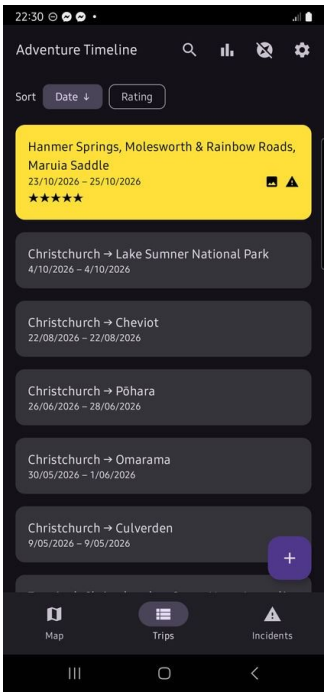
Finding 2: deleting a trip does not delete its media files (not fixed)

`deleteTrip` and `deleteAllTrips` in `TravelDiaryViewModel.kt` remove database rows only. `deleteMedia` (single item) does delete the file. After deleting the test trip its photo, video and voice-note files (about 27 MB) were still in the app's media folder; they were removed by hand. Real videos can be gigabytes, and

the confirmation says “This removes its legs, media and notes”. Suggested fix: delete the trip's app-owned media files (except ones attached to incidents that survive) when the trip is deleted.



Delete confirmation



Trip list after deletion

Finding 3: rotating the phone loses typed text and a voice recording in progress (fixed)

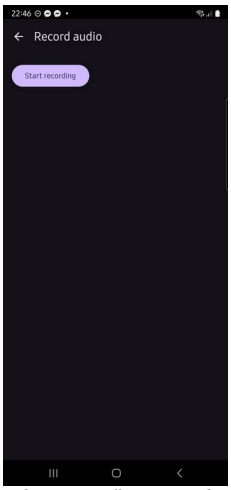
What happened. With the screen rotated to landscape and back, the New trip form came back empty (Title and Notes gone, Save disabled). The same happened in Add leg, the incident form and the Share message, and the Theme dialog closed. Rotating during a voice recording reset the screen to “Start recording” while the recorder kept running; the partial file was saved but not attached to any trip, so that voice note was effectively lost. This matters for a phone in a bike mount.

Cause. `MainActivity` did not handle configuration changes, so Android recreated it on rotation and the screens' plain `remember` state was lost.

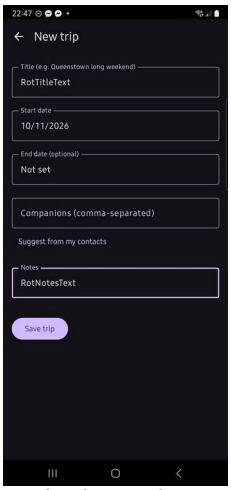
Fix. `android:configChanges="orientation|screenSize|smallestScreenSize|screenLayout|keyboard|keyboardHidden"` on `MainActivity` in `AndroidManifest.xml`, so rotation no longer recreates the activity. Re-tested with the same steps: forms, the Share message and the dialog keep their state, and a recording continues and saves. Not covered: the same text is still lost if Android kills the app in the background; the forms would need `rememberSaveable` for that.



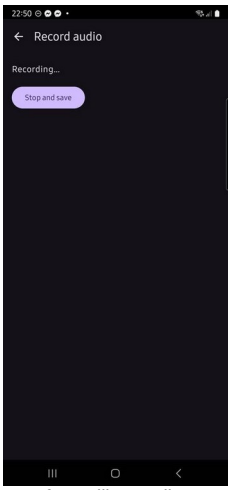
Before: New trip form with text, landscape, then back in portrait with the fields empty.



Before: recording reset after rotation.



After: form text kept.



After: still recording.

Other observations

- The Share tab shows an empty black map area for a trip with no route; the Map tab says “No route yet”.

- GPX and KML exports of a trip with no route are valid but empty, with no warning.
- I found no way to delete a companion in the app. The test companion “Test Buddy” remains as one extra row (see Final state).
- Camera preview and review screens are in portrait; one transient landscape state was seen after the camera returned.

Not tested

Feature	Why
Free up space	The action can replace duplicate media copies with links; an automated safety check blocked the tap, so it was left alone.
Delete all trips	Destructive on real data; deliberately not run.
Incident-level camera/video capture	Same code pattern as Finding 1 but not exercised.
Map tiles for areas never viewed online	Only cached tiles display offline; not tested in an uncached area.
Restore from an older backup, Timeline import with a real file, contacts-granted linking	Not run, to avoid changing real data further.

Final state

- Diary matches the starting snapshot (21 trips, 395 legs, 7 media items, 1 incident, 6,867 route points) except one extra companion row, “Test Buddy”. I found no way to delete it in the app; a direct database edit was attempted but blocked by a safety check, so nothing was changed. It can be removed by hand.
- My test backup (3.7 GB) and test media files were deleted. The owner’s own 9 Nov backup was not touched.
- Wi-Fi on; auto-rotate on; developer option “destroy activities” off; camera access granted “Only this time”.
- The Finding 1 and Finding 3 fixes are in the working tree and not committed.
- The second throwaway trip, “ZZ Rotation Test”, and its photo, video and voice note were removed through the app (media first, so the files were deleted, then the trip). The trip list is back to the owner’s trips only.

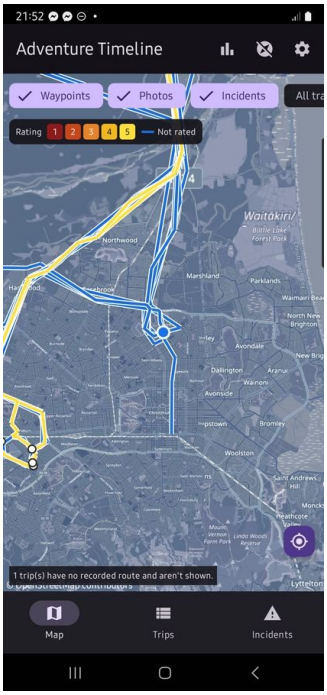
Screenshots

These include the owner’s real map, trip list and stats (first group). Everything else is the throwaway test trip.

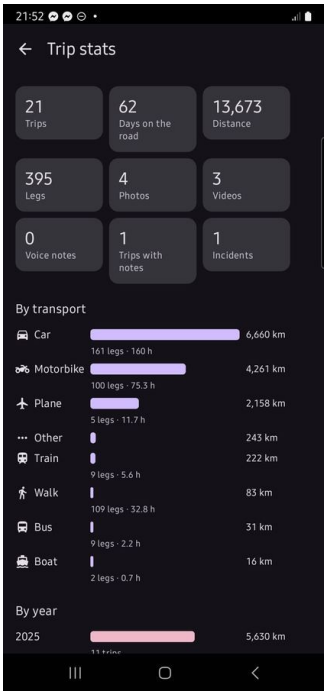
Start-up, map and lists



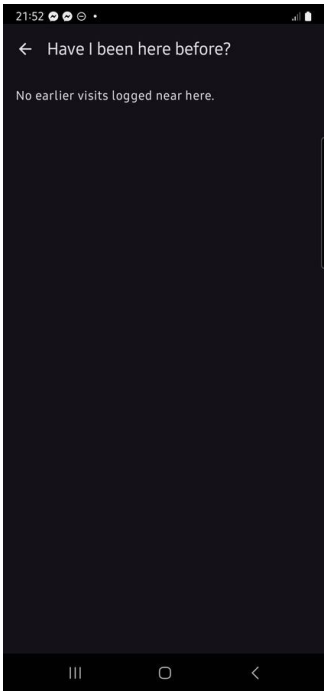
All-trips map, offline (real data)



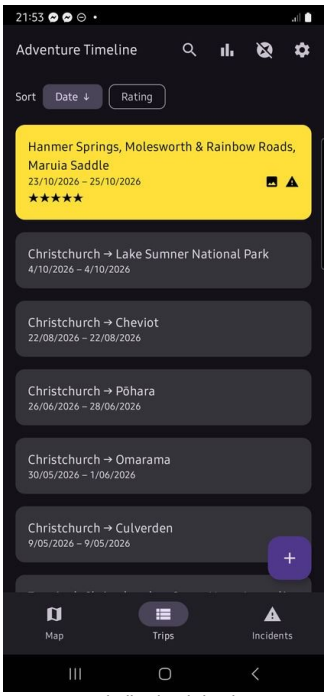
Centre on my location



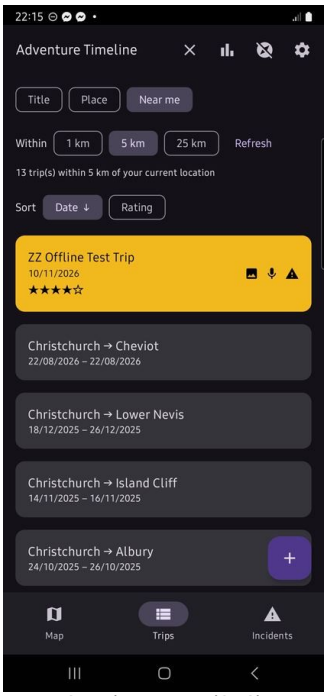
Trip stats (real data)



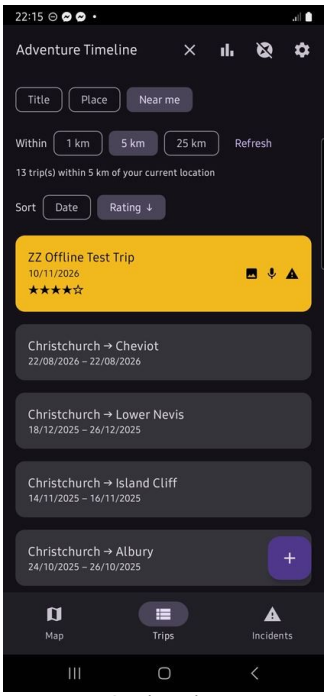
Have I been here before?



Trip list (real data)

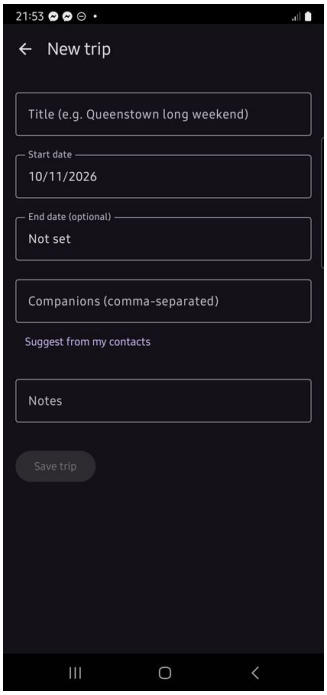


Search: Near me (GPS)

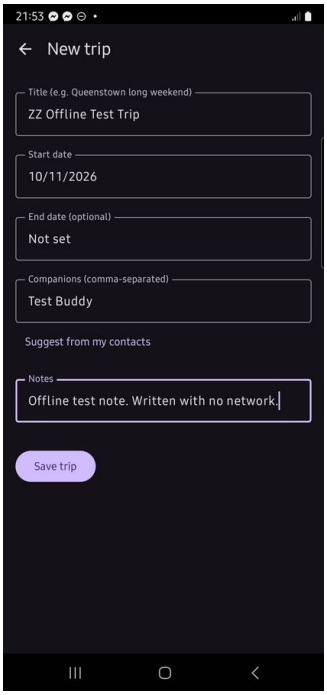


Sort by rating

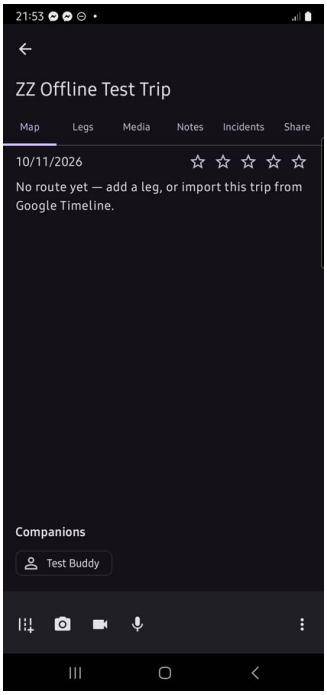
Create and edit a trip



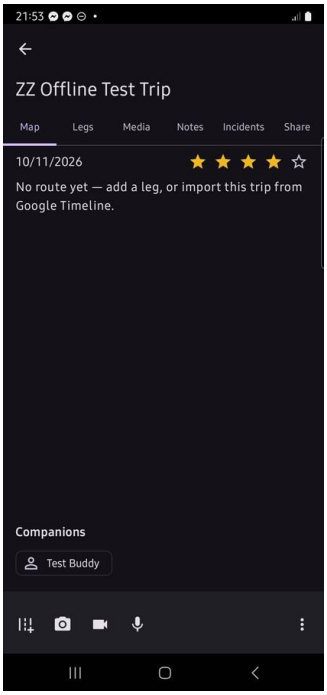
New trip



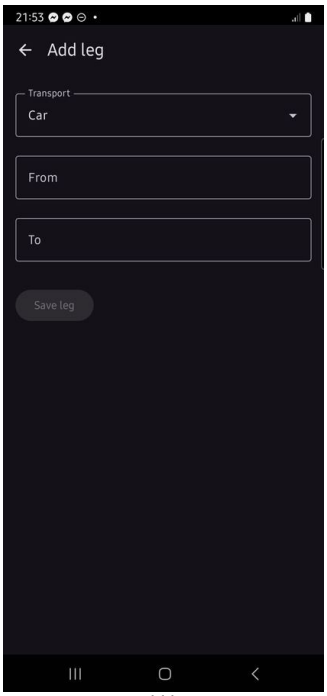
Filled in



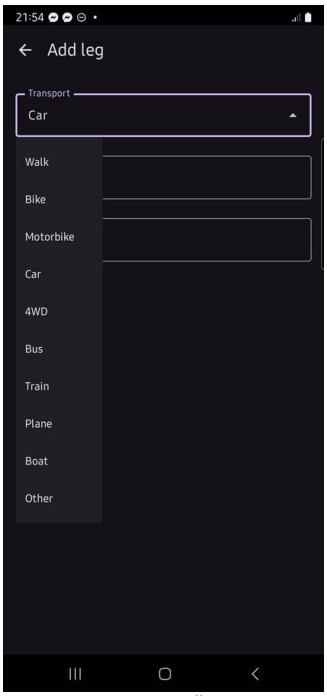
Saved



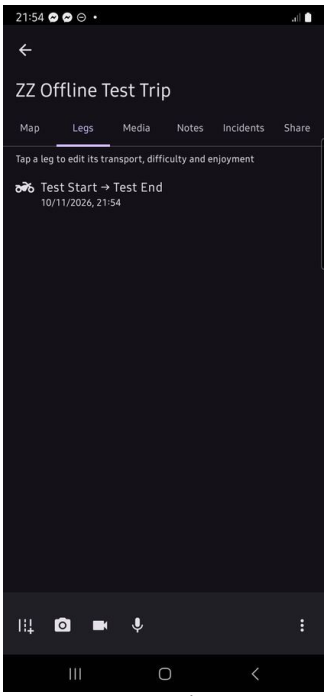
Rated 4 stars



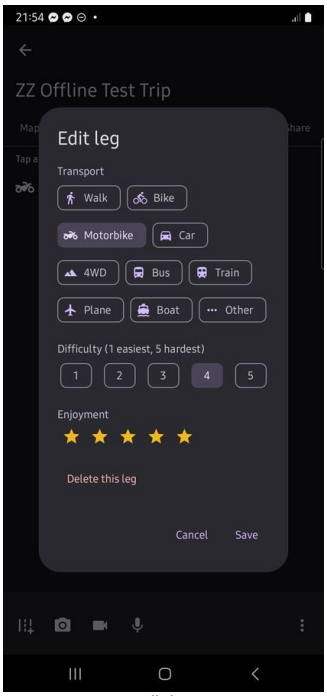
Add leg



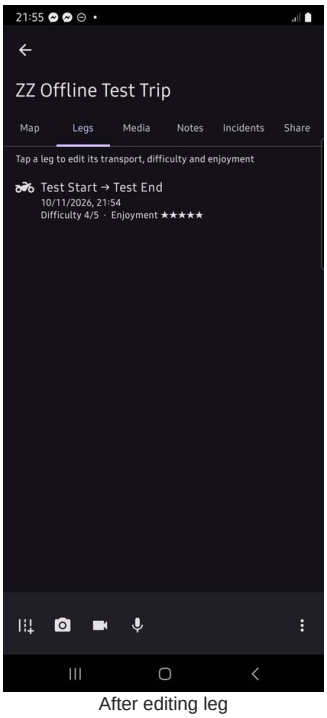
Transport list



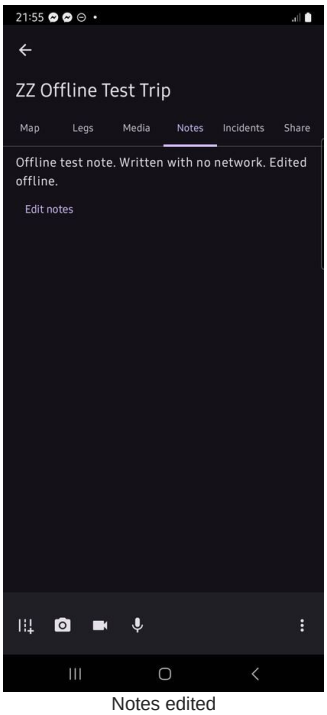
Legs tab



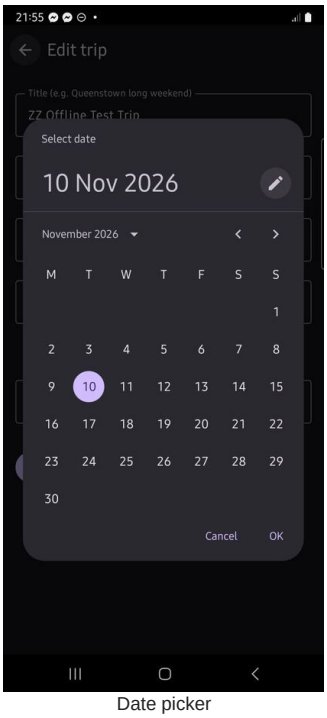
Edit leg



After editing leg

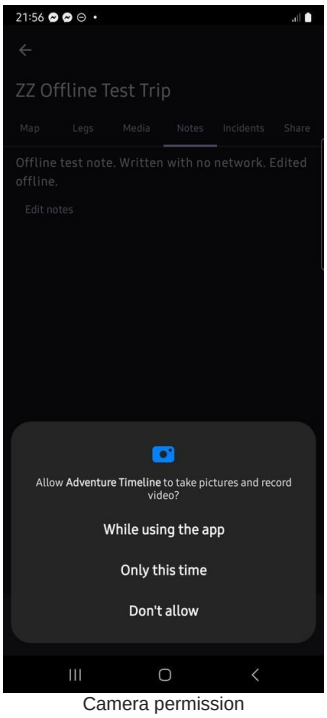


Notes edited

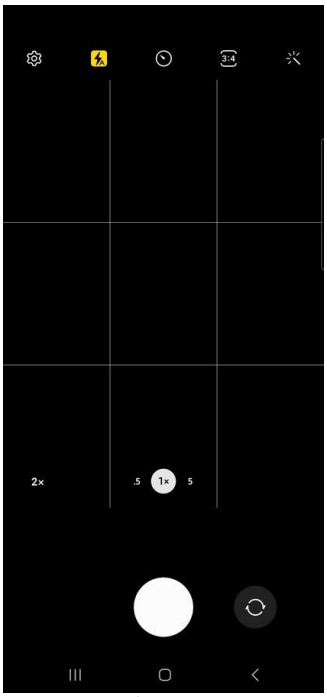


Date picker

Camera, video and voice notes



Camera permission



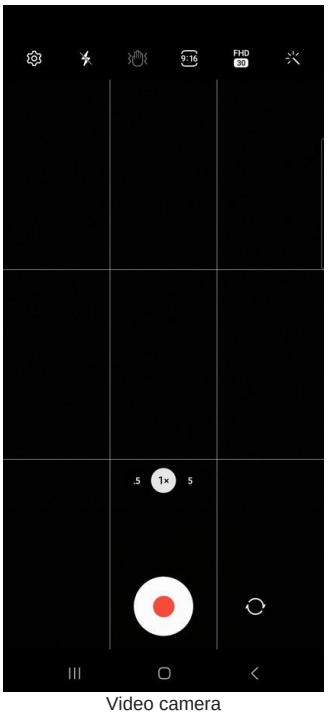
Camera



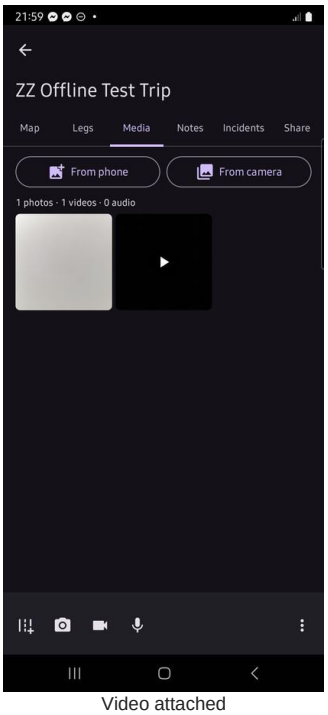
Photo review



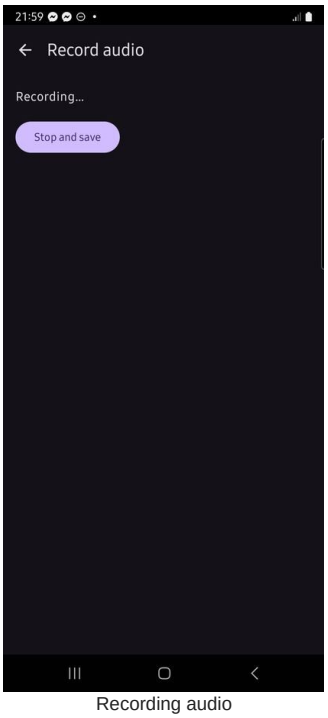
Photo attached



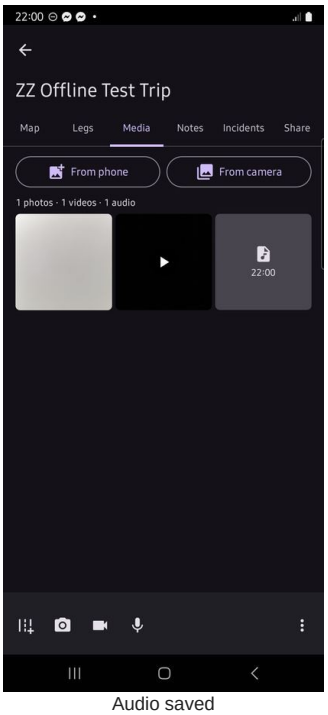
Video camera



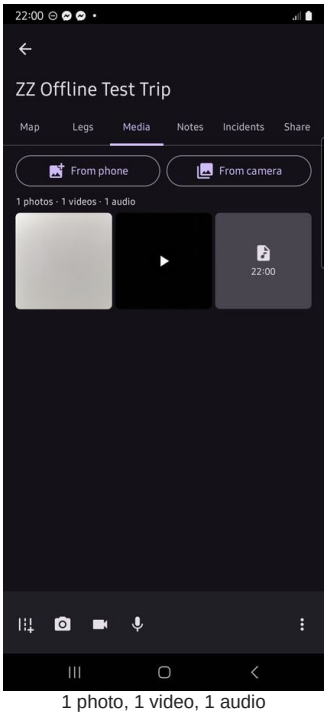
Video attached



Recording audio



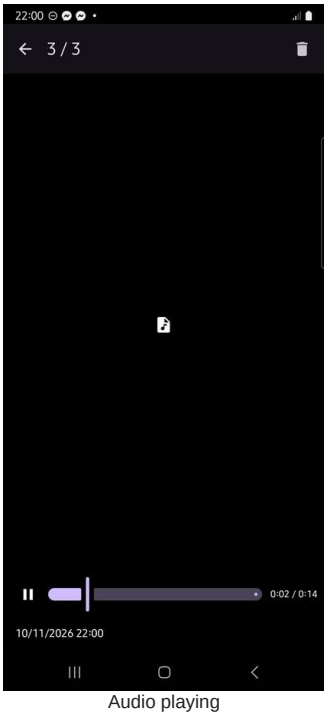
Audio saved



1 photo, 1 video, 1 audio

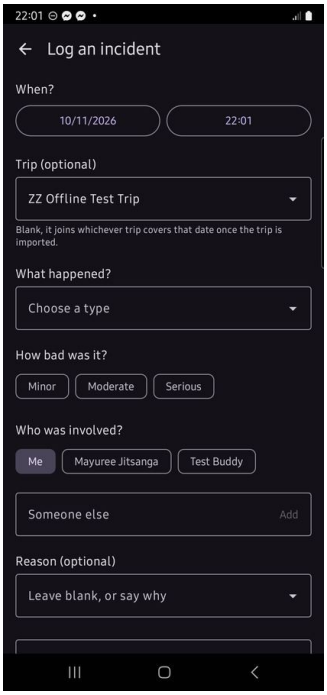


Viewer: coordinates fallback

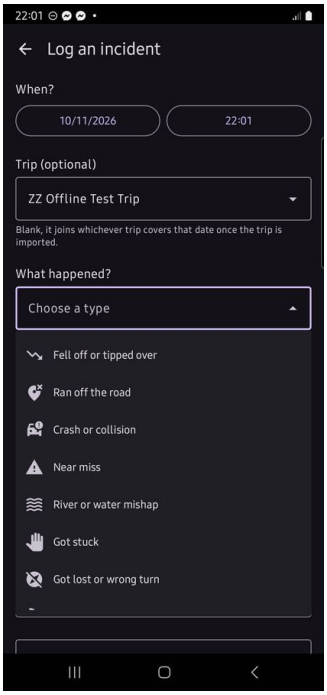


Audio playing

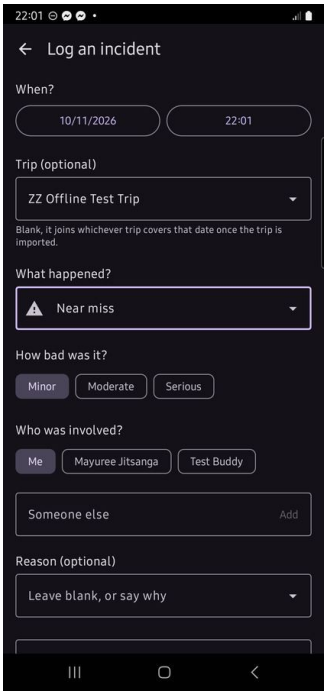
Incidents



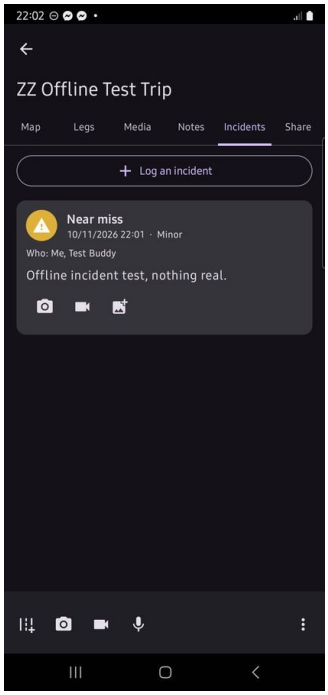
Log an incident



Incident types

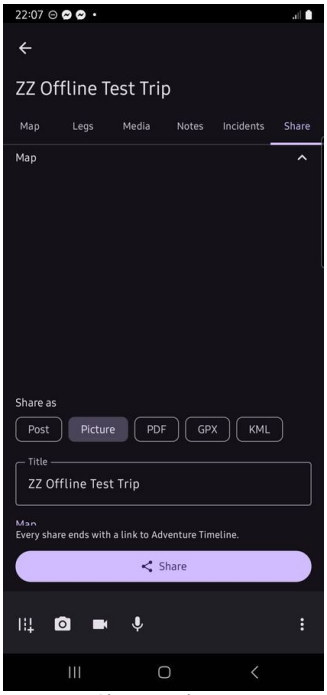


Filled in

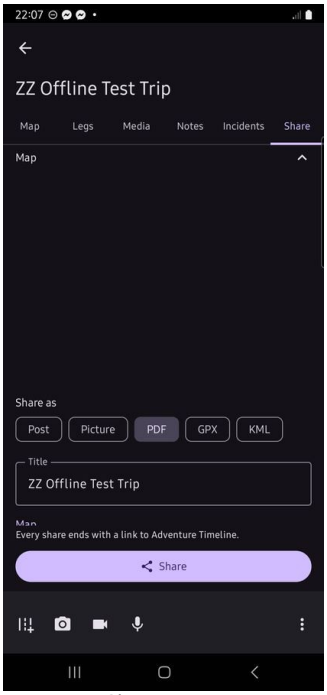


Saved on the trip

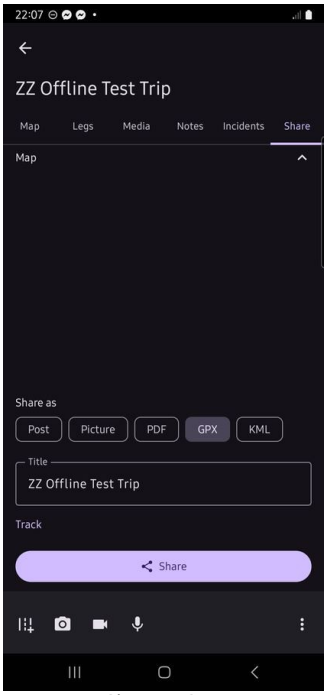
Sharing and export



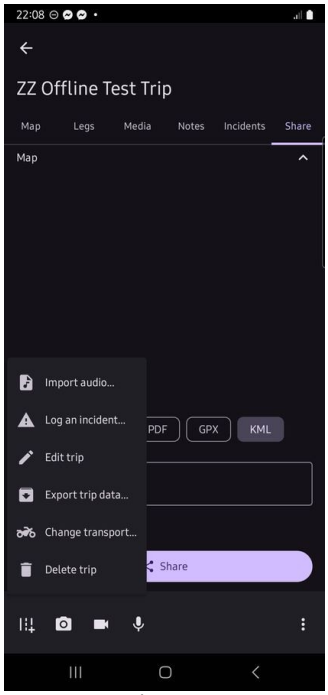
Share as Picture



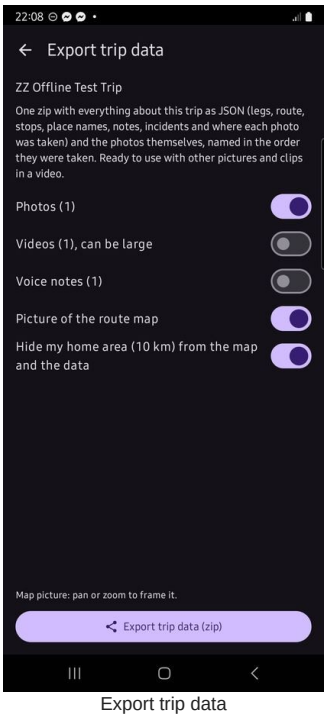
Share as PDF



Share as GPX

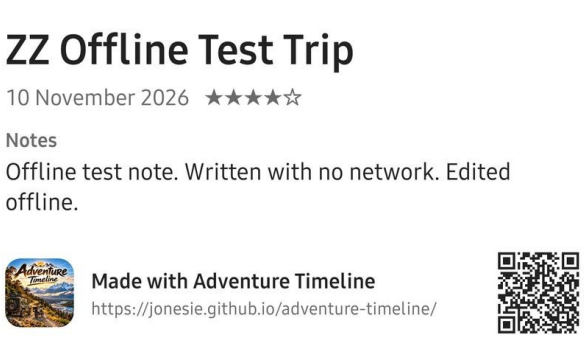


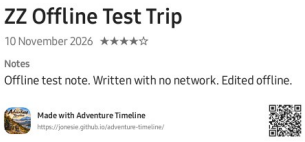
Trip menu



Export trip data

Generated Picture export and the first page of the generated PDF (both built offline):

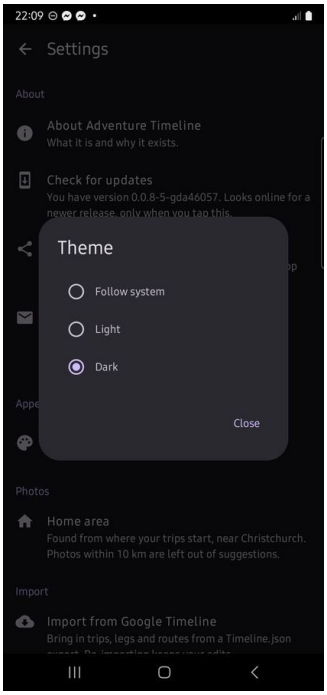




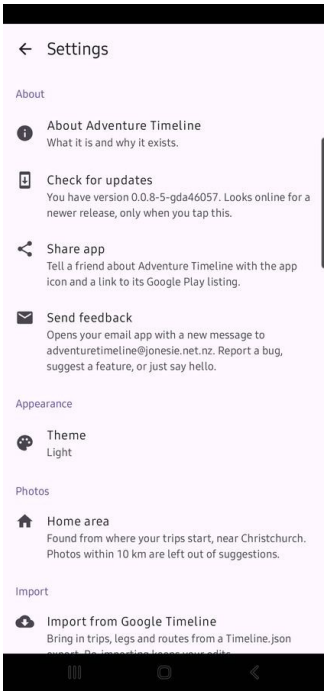
Rotation (after the fix, landscape view of the New trip form)



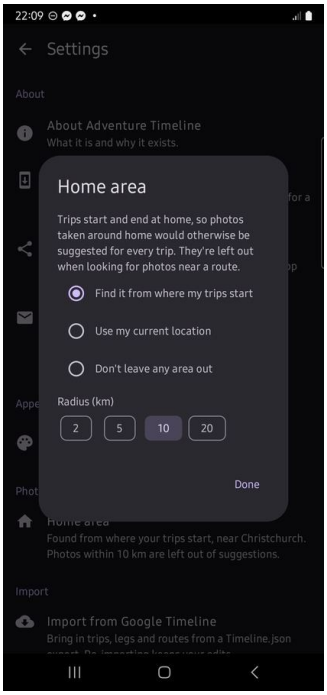
Settings, backup and restore



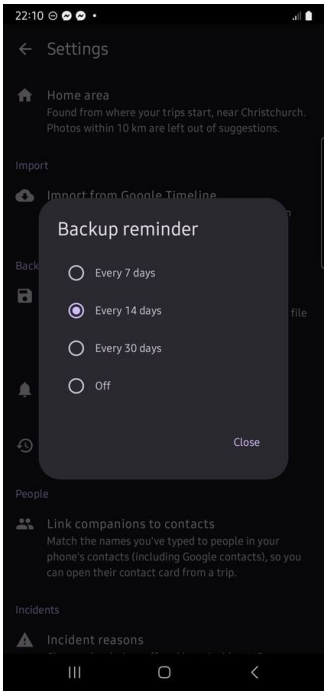
Theme



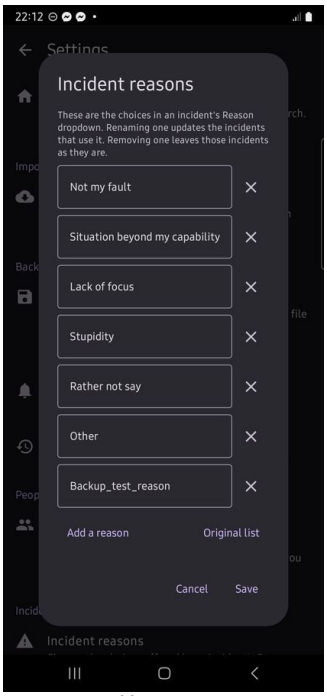
Light theme



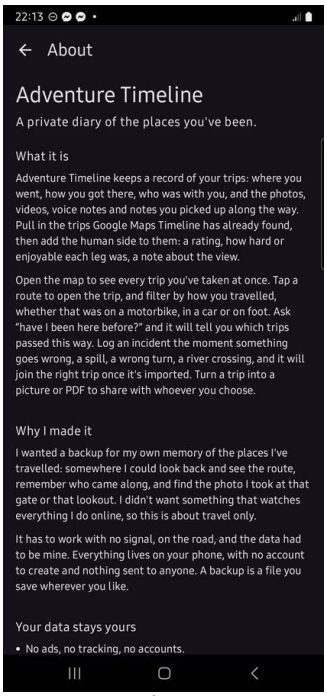
Home area



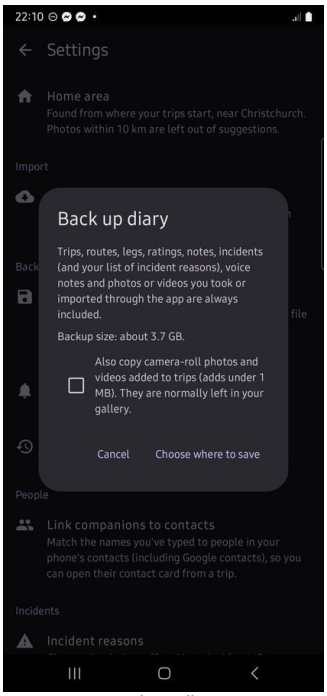
Backup reminder



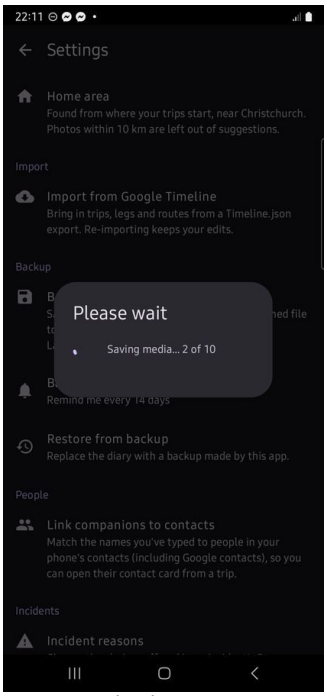
Incident reasons



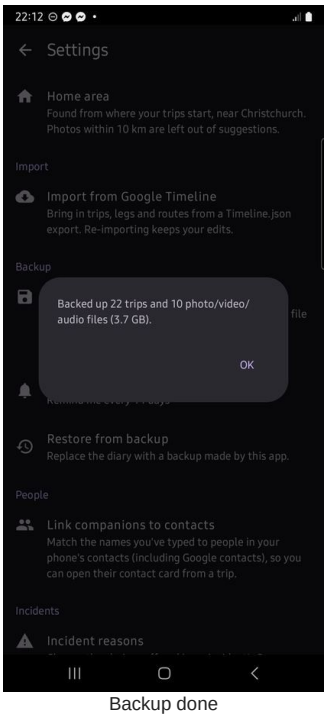
About



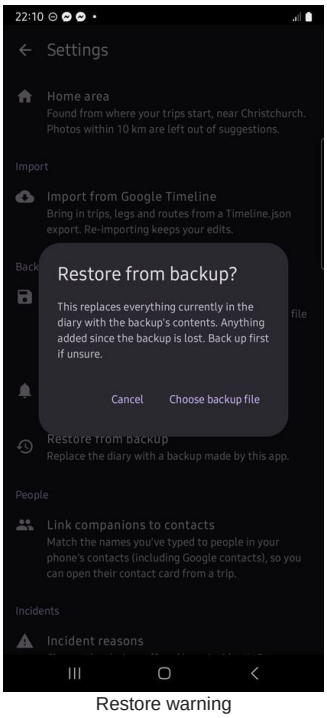
Back up diary



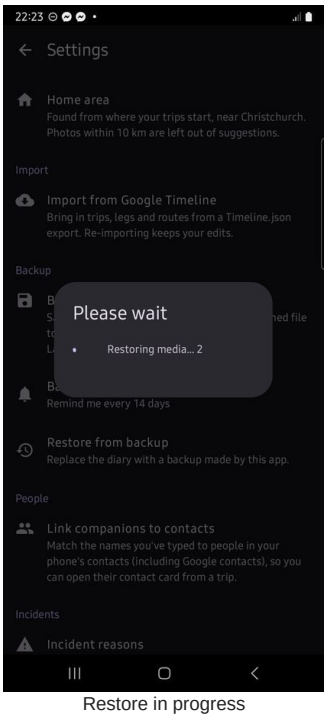
Backup in progress



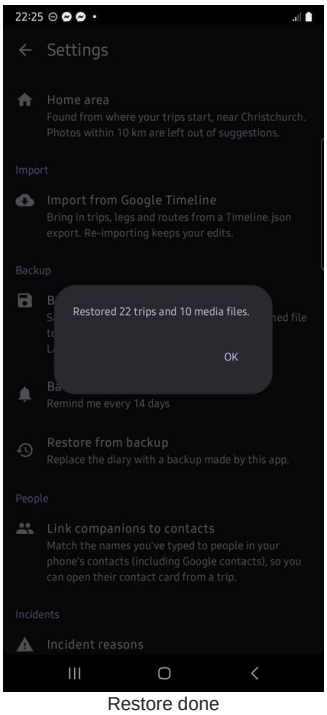
Backup done



Restore warning

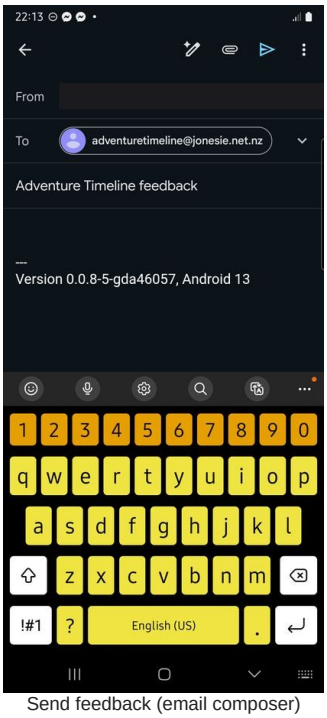


Restore in progress

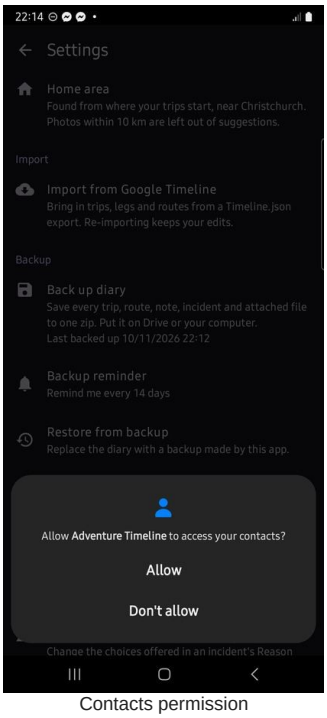


Restore done

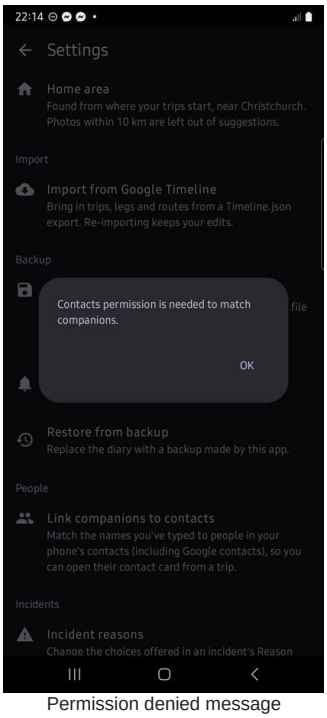
Feedback, contacts, import and network features



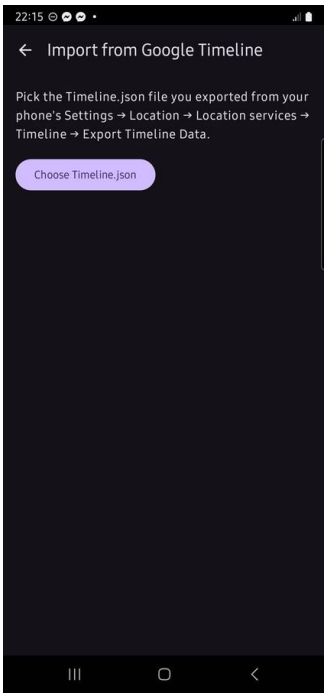
Send feedback (email composer)



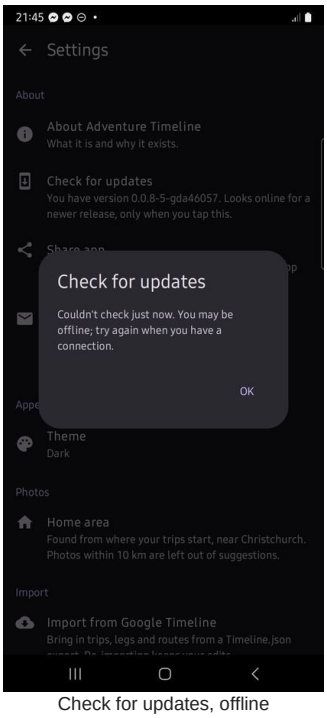
Contacts permission



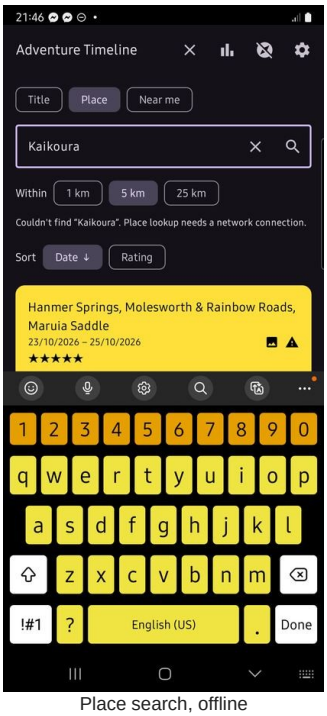
Permission denied message



Import screen



Check for updates, offline



Place search, offline